

when equipment faults are less likely or for end-of-life when faults are not only likely but potentially more detrimental?

The preferred approach is an in-system programmable sensor, which allows reconfigurability to changes in life cycle. For instance, infrequent monitoring for the lowest power consumption during early life, followed by reconfiguration to frequent (user-programmed period) monitoring once a shift (warning threshold) has been observed; in addition to the continuous monitoring-for and interrupt-driven notification of user-programmed alarm thresholds.

Identifying shifts/trends

The discussion on adapting the sensor to changes in equipment life-cycle is somewhat dependent on knowledge of a baseline equipment response. Even simple analogue sensors can allow this assuming the operator takes measurements; carries out the off-line analysis; and stores this data off-line with proper tagging to the specific equipment and probe location. A preferred and less error-prone approach would allow baseline FFT storage at the sensor head, thus eliminating any potential for misplaced data. The baseline data also helps in establishing alarm levels, which again would ideally be programmed directly at the sensor. Thus in any subsequent data analysis/capture, where warning or fault conditions are detected, a real-time interrupt can be generated.

Documentation/traceability

Within a factory setting, a proper vibration analysis program may be monitoring tens or even hundreds of locations, whether by handheld probe or embedded sensor. Over the course of a given piece of equipment's lifetime, this may produce the need for capturing thousands of records. The integrity of the predictive maintenance program depends on proper mapping to location and time of the sensor collection point. For lowest risk and the most valuable data, the sensor should have a unique serial number. It should also have the ability to time-stamp the data in addition to embedded storage.

Reliability

What if the sensor becomes faulty (performance shift), rather than the equipment? Or, if operating with a fully autonomous sensor (as described as the ideal), how confident can we be that the sensor continues to work at all? With many existing transducers such as piezo-based, these situations present a serious limitation because simple piezo sensors have no means of providing an in-system self-test.

There is always a lack of confidence in the consistency of data recorded over time, and in the end-of-life critical monitoring phase, where real-time fault notification is time- and cost-critical and can be a significant safety concern. There is always a concern that the sensor could become non-functional. An essential requirement of a high-confidence predictive maintenance program is the ability to remotely self-test the





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transducer. Fortunately this is possible with some MEMS-based sensors. An embedded digital self-test capability will close the final gap on a reliable vibration monitoring system.

The intricacies of vibration monitoring, particularly capturing accurate representations of the vibration profile and then correctly interpreting the data are highly complex disciplines. For many who are wishing to implement vibration monitoring, the optimum solution lies far beyond the transducer element. Much of the complexity lies in the data analysis, where a typical time-based analysis of the equipment produces a complex waveform combining multiple error sources and providing little discernible information prior to FFT analysis.

Most piezo-based sensor solutions rely on external computation and analysis of the FFT. This approach eliminates the possibility of real-time notification and greatly increases the design burden on the equipment developer.

With the high level of integration and a simplified programmable interface, these sensors enable easier adoption of vibration sensing, previously limited to a handful of highly skilled technologists with decades of analytical experience in machine vibration. ●

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